Date:________ Day:________

1 Write a Program that display a message and value of integer and character Variables:-

```
#include<iostream>
using namespace std;
int main()
{
    int n=10;
    char ch='*'
    cout<<"Testing Output...";
    cout<<n;
    cout<<ch;
}
```

Output

Testing Output...10*

2 Write a Program add two floating Number and display Sum:-

```
#include<iostream>
using namespace std;
int main()
{
    float a,b;
    a=24.27;
    b=41.50;
    Cout<<a<<"+"<<b<<"="<<a+b;
}
```

output

24.27+41.50=65.770004

3 Calculate Area:-

```
#include<iostream>
using namespace std;
int main()
{
    int H,W,A;
    H=5;
    W=4;
    A=H*W;
    cout<<"Area of Square ="<<A;
}
```

Output

Area of Square=20

4 Display Pattern:-

```
#include<iostream>
using namespace std;
int main()
{
    cout<<"*"\n**\n***\n ****";
}
```

Output

```
*
**
***
****
```

5 Write a Program that explain the use a (Setw) manipulator:-

Date:________ Day:________

```
#include<iostream>
#include<iomanip>
using namespace std;
int main()
{
    int n=3928;
    double d=91.5;
    char str[]="OOP using C++";
    cout<<"("<<setw(5)<<n<<")"<<endl;
    cout<<"("<<setw(8)<<d<<")"<<endl;
    cout<<"("<<setw(16)<<str<<")"<<endl;
}
```

output

```
3928
    91.5
   OOP using C++
```

6) **Using setprecision manipulator:-**

```
#include<iostream>
#include<iomanip>
using namespace std;
int main()
{
    double r,n1=132.364,n2=26.91;
    r=n1/n2;
    cout<<r<<endl;
    cout<<setprecision(5)<<r<<endl;
    cout<<setprecision(4)<<r<<endl;
```

output

```
4.918766
4.91877
4.9188
4.919
4.92
4.9
```

Date:

```
cout<<setprecision(3)<<x<<endl;
cout<<setprecision(2)<<x<<endl;
cout<<setprecision(1)<<x<<endl;
}
```

7) Use showpoint manipulator:-

```
#include<iostream>
#include<iomanip>
using namespace std;
int main()
{
  double x,y,z;
  x=15.674;
  y=235.73;
  z=9525.9864;
  cout.setf(ios::fixed, ios::floatfield);
  cout.setf(ios::showpoint);
  cout<< setprecision(2) <<"setprecision(2)"<<endl;
  cout<<"x="<< x<<endl;
  cout<<"y="<<y<<endl<<"z="<<z<<endl;
  cout<<setprecision(3) <<"setprecision(3)"<<endl;
  cout<<"x="<<x<<endl<<"y="<<y<<endl<<
  "z="<<z<<endl;
```

output

```
Setprecision(2)
x=15.67
y=235.73
z=9525.99
setprecision(3)
x=15.674
y=235.730
z=9525.986
Setprecision(4)
x=15.6740
y=235.7300
z=9525.9864
15.674  235.730  9525.986
```

Date: Day:

```
    cout<<setprecision(4)<<"setprecision(4)"<<endl;
    cout<<"x="<<x<<endl<<"y="<<y<<endl<<"z="
       <<z<<endl;
    cout<<setprecision(3)<<x<<" ";
    cout<<setprecision(2)<<y<<" "<<setprecision(4)
       <<z<<endl;
}
```

8 **Use setfill manipulator:-**

output
*******OOP using C++
@@@@@@@ OOP using C++
======== OOP using C++

```
#include<iostream>
#include<iomanip>
using namespace std;
int main()
{
    char str[]="OOP using C++";
    cout<<setw(20)<<setfill('*')<<str<<endl;
    cout<<setw(20)<<setfill('@')<<str<<endl;
    cout<<setw(20)<<setfill('=')<<str<<endl;
}
```

9) **Input Name, Age, address and display its value:-**

```
#include<iostream>
using namespace std;
int main()
{
```

Date: Day:

```
char name[25], city[30];
int age;
cout<<"Enter your age:";   cin>>age;
cout<<"Enter your Name:";
cin >> name;
cout<<"Enter your city:";
cin>> city;
cout<<"\n Your Name:"<<endl;
cout<<"Your city:"<<endl;
cout<<"Your age:"<<endl;
}
```

Output

Enter your age: 16
Enter your Name: Ali
Enter your city: Lahore

Your Name = Ali
Your city: Lahore
Your age: 16

10) **calculate Intrest Input amount, rate of intrest and the number of years display it**

Date: ______ Day: ____

```
#include<iostream>
using namespace std;
int main()
{
    double p,r,t,i;
    cout<<"Enter amount,rate,time:";
    cin>>p>>r>>t;
    i=(p*r*t)/100;
    cout<<"\n Principle Amt="<<p;
    cout<<"\n Rate="<<r<<"%";
    cout<<"\n Time="<<t<<"yrs";
    cout<<"\n Intrest Amt="<<i;
}
```

Output

Enter amount,rate,time 1000 5 3
Principle Amt= 1000
Rate= 5%
Time= 3yrs
Intrest Amt= 150

11 **Input character display ASCII value**

```
#include <iostream>
#include <conio>
using namespace std;
```

Date: Day:

```
int main()
{
    char ch;
    cout<<"Enter a Character";
    cin>>ch;
    int n=ch;
    cout<<"The ASCII for"<<ch<<"is"
    <<n<<"."<<endl
}
```

Output

```
Ente a character a
The ASCII fo a is.97
```

12 **Input dividen & divider and Display Quotient & reminder:-**

```
#include<iostream>
using namespace std;
int main()
{
    int div,dis,q,r;
    cout<<"Enter dividen and divisor:";
    cin>>div>>dis;
    q=div/dis;
    r=div%dis;
```

Date: Day:

```
    cout<<"Quotient="<<q<<endl;
    cout<<"Reminder="<<r;
}
```

Output

Enter dividend & divisor: 20 3
Quotient= 6
Reminder= 2

13 **Input two Numbers and Swap it:-**

```
#include<iostream>
using namespace std;
int main()
{
    int a,b,temp
    cout<<"Enter first number:";
    cin>>a;
    cout<<"Enter the second number:";
    cin>>b;
    cout<<"You Input the numbers as"<<a
    <<"and"<<b<<endl;
    temp=a;
    a=b;
    b=temp;
    cout<<"After Swap:"<<a<<"and"<<b<<endl;
}
```

Date: Day:

Output:-

Enter first number: 10
Enter Second number: 20
You Input the number as 10 and 20
After swap 20 and 10

14 **Swap Numbers without using third variable:-**

```
#include<iostream>
using namespace std;
int main()
{
    int x,y;
    cout<<"Enter 2 integers respectively:";
    cin>>x>>y;
    cout<<"The original is x="<<x<<"and y="<<y;
    x=x+y;
    y=x-y;
    x=x-y;
    cout<<"After swap x="<<x<<"and y="<<y;
}
```

Output:-

Enter ~~t~~ 2 integars respectively 2 3
The original is x=2 and y=3
After Swap x=3 and y=2

Date:________ Day:________

15) Input distance, speed Calculate time, and display it:

```
#include <iostream>
using namespace std;
int main()
{
    double d,t,s;
    cout<<"Enter distance in miles:";
    cin>>d;
    cout<<"Enter ~~time~~ speed of vchical (mph):";
    cin>>s;
    t= d/s;
    cout<<"Time is ="<<t<<"hours"<<endl;
}
```

Output:

```
Enter distance in miles:100
Enter speed of vchical (mph):10
Time is = 10 hours
```

16) Enter base, hight and calculate Area:-

```
#include <iostream>
using namespace std;
int main()
{
```

Date:______ Day:______

```
    float b,h;
    double A;
    cout<<"Enter base:"<<b;
    cout<<"Enter height:"<<h;
    A=0.5*b*h;
    cout<<"Area="<<area;
}
```

Output

```
Enter base: 10.5
Enter height: 5.4
Area= 28.35
```

17 **Enter seconds & convert into hh-mm-ss format:-**

```
#include<iostream>
using namespace std;
int main()
{
    int sec,s,m,h;
    cout<<"\nEnter time in seconds:";
    cin>>sec;
    h=sec/3600;
    m=sec/60;
    s=sec%60;
    cout<<"\nHH-MM-SS="<<h<<":"<<m<<":"<<s;
}
```

Date: Day:

Output:-

Enter time in Seconds: 5300

HH-MM-SS= 1:28:20

18 **Enter temperature in Celcius convert into Fahrenhiet:-**

```
#include<iostream>
using namespace std;
int main()
{
    float c,f;
    cout<<"Enter temperature in Celsius:"
    cin >> c;
    f=9/5*c+32;
    cout<<"Temperature in Fahrenhiet:";
}
```

Output:-

Ente Temperature in celsius 15.50

Tempratuwre in Fahrenhiet: 59.9

19 **Enter height in Inches and convert in cm:-**

```
#include<iostream>
using namespace std;
int main()
```

Date:

```
{
    int h;
    float hc;
    cout<<"Enter height in inches:";
    cin>>h;
    ~~cout~~ hc= ha*2.54;
    cout<<"Your height in centimeters is:"
}
```

Output

Enter height in inches: 20

Your height in centimeters is: 50.8

20 **Enter Radius calculate Area and circumference:-**

```
#include<iostream>
using namespace std;
int main()
{
    float a,r,c;
    cout<<"Enter radius="
    cin>>radius;
    a=r*r*3.141;
    c=2*3.141*r;
    cout<<"Area is:"<<a<<endl;
    cout<<"circumference:<<c<<endl;
}
```

CS CamScanner

Date:________ Day:________

Output:-

Enter radius: 5

Area: 78.53

circumference: 31.41

21 **calculate final velocity by taking vi, acceleration, time, vf=vi+at**

```
#include<iostream>
using namespace std;
int main()
{
    int vi,a,t,vf;
    cout<<"Enter inital velocity:";
    cin>>vi;
    cout<<"Enter acceleration:";
    cin>>a;
    cout<<"Enter time:";
    cin>>t;
    vf=vi+a*t;
    cout<<"The final velocity is"<<vf<<endl;
}
```

Output:-

Enter the initial velocity: 20

Enter acceleration: 10

Date: Day:

Enter time: 5
The final velocity is: 70

22 Enter three numbers and swap it:-

```
#include<iostream>
using namespace std;
int main()
{
    int a,b,c;
    cout<<"Enter 3 digits:";
    cin>>a;
    b=a/100;
    a=a%100;
    c=a/10;
    a=a%10;
    cout<<"After Swap:"<<a<<b<<c;
}
```

Output:-

Enter 3 digits: 321
After Swap:123

23 Enter five numbers & swap it:-

```
#include<iostream>
using namespace std;
int main()
```

Date: Day:

```
{
    long int n,b,c,a,d;
    cout<<"Enter 5 digits:"<<;
    cin>>n;
    a=n/10000
    n=n%10000;
    b=n/1000;
    n=n%1000;
    c=n/100;
    n=n%100;
    d=n/10;
    n=n%10;
    cout<<"After Swap:"<<n<<d<<c<<b<<a;
}
```

Output:-

Enter 5 digits: 92174

After Swap: 47129

24 **Input even and odd number (X) even by(5) odd by(3) add both and (-) the result by1000:-**

```
#include<iostream>
using namespace std;
int main()
{
```

Date: Day:

```
int ev,od,r;
cout<<"Enter an even number:";
cin>>ev;
cout<<"Enter an odd number:";
cin>>od;
r=1000-((ev*5)+(od*3));
cout<<"Difference="<<r;
}
```

Output:-

```
Enter an even number:10
Enter an odd number:5
Difference=935
```

25 **Generate output:-**

10

20

19

```
#include<iostream>
using namespace std;
int main()
{
    int n=10;
    cout<<n<<endl;
```

Date: Day:

```
    n *=2;
    cout<<n<<endl;
    cout<<--n;
}
```

26 Enter hours converts into weeks, days, hours:-

```
#include<iostream>
using namespace std;
int main()
{
    int hrs,w,d;
    cout<<"Enter hours:";
    cin>>hrs;
    w=hrs/168;
    hrs=hrs%168;
    d=hrs/24
    hrs=hrs%24
    cout<<"weeks:"<<w<<endl;
    cout<<"Days:"<<d<<endl;
    cout<<"Hours:"<<hrs<<endl;
}
```

output

Enter hours: 200

Weeks: 1

Days: 1

Hours: 8

Date: Day:

27 **Enter rate of electricity, gas, petrol and increase 10% and Display:-**

```
#include<iostream>
using namespace std;
int main( )
{
    float e1,e2,g1,g2,p1,p2;
    cout<<"Enter electricity rate:";
    cin>>e1;
    cout<<"Enter petrol rate:";
    cin>>p1;
    cout<<"Enter gas rate:";
    cin>>g1;
     e2=e1*1.1
     p2=p1*1.1
     g2=g1*1.1
     cout<<"New electricity rate:"<<e2<<endl;
     cout<<"New Petrol rate:"<<p2<<endl;
     cout<<"New Gas rate:"<<g2<<endl;
}}
```

Output:-

Enter electricity rate: 11

Enter petrol rate : 69

Date:

Day:

Date:

Enter gas rate: 8

New electricity rate: 12.1

New Petrol rate: 75.9

New Gas rate: 8.8

28 Print output:

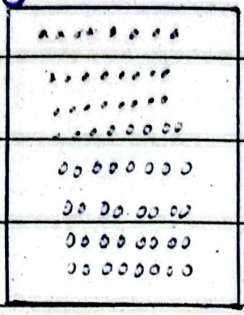
C
OM
PUT
ERIS
AWORL
DOFS
CIE
NC
E

```
#include<iostream>
using namespace std;
int main()
{
    cout<<"C"<<endl;
    cout<<"OM"<<endl;
    cout<<"PUT"<<endl;
    cout<<"ERIS"<<endl;
    cout<<"AWORL"<<endl;
```

Date: Day:

```
cout<<"DOFS"<<endl;
cout<<"CIE"<<endl;
cout<<"NC"<<endl;
cout<<"E";
}
```

29 Display



```
#include<iostream>
using namespace std;
int main()
{
   cout<<"........"<<endl;
   cout<<"........"<<endl;
   cout<<"........"<<endl;
   cout<<"........"<<endl;
   cout<<"........"<<endl;
   cout<<"........"<<endl;
   cout<<"........"<<endl;
   cout<<"........";
}
```